

Cautionary notes on the use of pedotransfer functions for estimating soil hydraulic properties

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Abstract

The performance of published pedotransfer functions was evaluated in terms of predicted soil water content, pressure heads, and drainage fluxes for a layered profile. The pedotransfer functions developed by Vereecken et al. (1989), Vereecken et al. (1990) were used to determine parameters of the soil hydraulic functions $\theta(h)$ and $K(h)$ which were then used as input to SWATRER, a transient one-dimensional finite difference soil water model with root uptake capability. The SWATRER model was used to simulate the hydraulic response of a multi-layered soil profile under natural climatic boundary conditions for a period of one year. The simulations were repeated by replacing the indirectly estimated water retention characteristic by (1) local-scale, and (2) field-scale mean observed $\theta(h)$ relationships. Soil moisture contents and pressure heads simulated at different depths in the soil profile were compared to measured values using these three different sets of hydraulic functions. Drainage fluxes at one meter below ground surface have also been simulated using the same three sets of hydraulic functions. Results show that simulations based on indirectly estimated moisture retention characteristics (obtained from pedotransfer functions) overpredict the observed moisture contents throughout the whole soil profile, but predict the pressure heads at shallow depths reasonably good. The results also show that the predicted drainage fluxes based on estimated retention functions are about four times as high compared to the drainage fluxes simulated using measured retention curves.

1. Introduction

The performance of water balance models is known to be very sensitive to the hydraulic parameters of the soil. Field and laboratory techniques for measuring the soil hydraulic functions, as reviewed by Bouma et al. (1983) and Klute and Dirksen (1986), remain relatively time consuming and costly. These authors, among others, showed that the meas-

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urement problem is further complicated because both physical and chemical soil properties exhibit significant temporal and spatial variabilities. As an approach to the measurement problem, indirect parameter estimation has been proposed (Bloemen, 1980; Rawls and Brakensiek, 1982; Haverkamp and Parlange, 1986; Wösten and van Genuchten, 1988; Vereecken et al., 1989; Van de Genachte et al., 1995). To overcome expensive in-situ or laboratory experiments for hydraulic properties, statistical models referred to as pedotransfer functions, PTFs, (Bouma, 1989) have been derived over the past decade (see van Genuchten et al., 1991 for a review). These PTFs estimate the soil moisture retention curves and/or the hydraulic conductivity relationship based on other readily available soil properties. Soil properties selected to indirectly estimate soil water retention are grouped into four categories: (1) soil particle size properties, (2) easily measurable hydraulic properties, (3) morphological properties, and (4) chemical properties. Different strategies have been developed to estimate the moisture retention characteristic (MRC) from these basic soil properties. For instance, Vereecken et al. (1989) applied a four-parameter retention function proposed by van Genuchten and Nielsen (1985) to describe the MRC for Belgian soils (for more details, the reader is referred to Vereecken et al., 1989):

$$S_c = \frac{1}{(1 + (\alpha|h|)^n)^m} \quad (1)$$

where

$$S_c = \frac{\theta - \theta_r}{\theta_s - \theta_r} \quad (2)$$

and S_c is the degree of saturation, θ is the volumetric water content (cm^3/cm^3), θ_r is the residual water content (cm^3/cm^3), and θ_s is the saturated water content (cm^3/cm^3). The parameters n and m define the shape of the curve, $1/\alpha$ is the air entry value and h is the pressure head (cm). Vereecken et al. used Eq. (1) assuming the parameter m was equal to 1, unlike the original van Genuchten formula with $m = 1 - 1/n$. The former form of the van Genuchten equation was accepted as adequate in generating the MRC and was therefore fitted to the entire data set of measured $\theta-h$ values. The estimated model parameters of the above equations were related to soil map information through multiple regression using two levels of input predictor variables. The low input level uses the sand, silt, and clay fraction, the carbon content, and the bulk density. The high input level contains more detailed information on the particle size distribution (nine fractions). Statistical analyses brought about the following regression equations for the low level inputs (see Vereecken et al., 1989):

$$\theta_s = 0.838 - 0.283(Bd) + 0.0013(Cl) [R^2 = 0.849] \quad (3a)$$

$$\theta_r = 0.015 + 0.005(Cl) + 0.014(C) [R^2 = 0.702] \quad (3b)$$

$$\log_c(\alpha) = -2.486 + 0.025(Sa) - 0.351(C) - 2.617(Bd) - 0.023(Cl) [R^2 = 0.621] \quad (3c)$$

$$\log_c(n) = 0.053 - 0.009(Sa) - 0.013(Cl) + 0.00015(Sa)^2 [R^2 = 0.556] \quad (3d)$$

where Bd is bulk density (g cm^{-3}), C is carbon content (%), Cl is clay content ($< 2\%$), and Sa is sand content (%). The coefficient of determination (R^2) for each statistical model is given between brackets. Logarithmic transformations for parameters α and n were required in order to obtain normally distributed dependent variables.

To complement previous work, Vereecken et al. (1990) presented a similar method for estimating unsaturated hydraulic conductivity from easily measured soil properties. The three-parameter Gardner (1958) equation best described the observed hydraulic conductivity for the given data set when compared to many other $K(h)$ models:

$$K = \frac{K_s}{1 + (b|h|)^{n'}} \quad (4)$$

where K is the unsaturated hydraulic conductivity (cm day^{-1}), K_s the saturated hydraulic conductivity (cm day^{-1}), b is a parameter inversely related to the air entry value (cm^{-1}), n' is a pore size index, and h is pressure head (cm). Regression equations for estimating the parameters in the Gardner model (Eq. 4) were established using similar soil properties as for the MRC parameters. For the lowest level of information, the derived regression equations are:

$$\log_e(K_s) = 20.62 - 0.96\log_e(Cl) - 0.66\log_e(Sa) - 0.46\log_e(C) - 8.43Bd \quad [R^2 = 0.200] \quad (5a)$$

$$\log_e(b) = -0.73 - 0.0187Sa + 0.058Cl \quad [R^2 = 0.315] \quad (5b)$$

$$\log_e(n') = 1.186 - 0.194\log_e(Cl) - 0.0489\log_e(Si) \quad [R^2 = 0.533] \quad (5c)$$

where Si is the silt content (%) and other predictor variables are as defined previously.

The development of PTFs for the estimation of soil hydraulic functions is of great significance to soil physicists and hydrologists, particularly in water balance studies and in quantified land evaluation at the regional scale (Bouma and van Lanen, 1987). Bouma (1989) states that "A major challenge for soil science is to 'translate' data we have to data we need, if only because there will not be funds available to obtain, for example, $K(\theta)$ and $\theta(h)$ data on a large scale". Soil hydraulic properties can be estimated using the large soil databases which contain information on basic soil properties and which are already available in many countries (e.g. Van Orshoven et al., 1988). For many applications at the regional scale, soil hydraulic properties are required but are simply not available and thus have to be estimated in an indirect way. Already a lot of work has been done to develop such estimation procedures and the large potentials and importance of this area of research have recently been addressed by van Genuchten et al. (1991).

Different methods in deriving parameters for soil hydraulic functions can be used, such as direct measurement combined with curve fitting or the use of PTFs. Comparisons of these methods are usually based on a statistical analysis of the hydraulic functions themselves (Baker and Bouma, 1976; Tietje and Tapken, 1993). However, soil hydraulic functions serve as input data for simulation models used in soil water balance studies. Therefore, if we want to evaluate different procedures for estimating hydraulic parameters we should base our decision on criteria or properties that are related to practical applications, such as water deficit in the growing season, downward water flux below a given depth, etc.

The goal of this study was to evaluate the effect of using indirectly estimated soil hydraulic functions for predicting the field water balance using the SWATRER model. The soil water retention characteristic and the hydraulic conductivity relationship were estimated from basic soil properties based on the pedotransfer functions of Vereecken et al. (1989), Vereecken et al. (1990). Simulations were also done with measured retention functions and their performance was also tested. Time series with field observations of water contents and pressure heads were compared with predicted values based on estimated and measured retention functions. Calculated drainage fluxes based on observed retention functions served as a reference in the comparison with drainage predicted using estimated retention functions. Since drainage data is usually difficult to obtain in field situations, unless tile drains are present, the least we can assess is the effect of using estimated retention functions on predicted drainage. The last criteria is of paramount importance in both agricultural and environmental issues, especially when contamination of the groundwater by infiltrating water of poor quality is of concern. However, evaluation of the performance of indirect estimation methods are seldom reported in the literature, and more field evidence on the effectiveness of these estimation methods should be established. This requirement was the motivation for this study.

2. Materials and methods

2.1. The SWATRER model

Traditional approach to modelling water flow in porous media is based on the one dimensional partial differential equation known as the Richards' equation, given as:

$$\frac{\partial h}{\partial t} = 1/C(h) \frac{\partial [K(h)(\partial h/\partial z + 1)]}{\partial z} + S(h)/C(h) \quad (6)$$

where h is soil water pressure (cm), $C(h) = d\theta/dh$ is the differential moisture capacity (cm^{-1}), z is the vertical coordinate directed positive upward (cm), $K(h)$ is the unsaturated hydraulic conductivity (cm day^{-1}), and S is a source/sink term. Eq. (6) is solved by a finite difference scheme as proposed by Haverkamp et al. (1977). The solution is an implicit numerical scheme with an explicit linearization of the differential moisture capacity. In this study the transient one-dimensional finite difference soil water root uptake model SWATRER (Dierckx et al., 1986) was used. The model applies a simple sink term and different types of boundary conditions (for instance free drainage, varying water table, etc.) at the bottom of the soil profile. Additional features in the program include the introduction of Ritchie's model (Ritchie, 1972) for calculating the actual evaporation and the curve number method developed by the US Soil Conservation Service (1972) for estimating runoff. Potential evapotranspiration is provided as input, to reduce the complexity of the input data.

A major problem in applying water balance models such as SWATRER concerns the lack of data on hydraulic properties for a particular soil. Soil hydraulic behaviour is characterized by the soil water retention curve which defines the volumetric water content as a function of the soil water pressure head, $\theta(h)$, and the hydraulic conductivity curve which relates the hydraulic conductivity to water content, $K(\theta)$, or pressure head, $K(h)$.

Table 1
Horizon description for the soil profile in the Neuenkirchen catchment

Horizon	Description
Mp, Ms	silty–loamy colluvial sediment with low organic matter content
Ah	humus horizon
ICv	calcareous loess

2.2. Field data

2.2.1. Catchment characteristics and sources of data

Validation of the aforementioned PTFs (Eqs. 3(a)–3(d) and 5(a)–5(c)) was done by comparing calculated water balance components (moisture content, pressure head) with time series of these variables measured at several sites in a small agricultural catchment. The selected catchment was located in Neuenkirchen, situated about 35 kilometers south of Braunschweig, Federal Republic of Germany. It is a small catchment drained by the Ohebach Creek with an area of approximately 1 km². The Braunschweig Research Group (BRG) made a coordinated effort to characterize the Neuenkirchen catchment in terms of soils, hydrology and crop response (Bork and Rohdenburg, 1986). Most of the important catchment data used in this study were taken from the BRG data set.

For our validation purposes, five profiles along a transect were selected based on the availability of the following measured data for each of the profiles: (1) general soil data such as description of layer and horizon, soil texture, humus content and other relevant soil information, (2) volumetric moisture content and pressure head, and (3) groundwater level observations near the test profile. Table 1 gives the horizon description of the soil profile discussed here and Fig. 1 shows the same representative soil profile together with the observed and estimated retention functions. For each of the five soil profiles, the soil water balance for a period of one year was calculated using SWATRER. In this paper we only show results for one profile (notably profile STN 3539) because the hydraulic behaviour in the remaining profiles was similar.

2.2.2. Estimating soil hydraulic functions for the selected soil profiles

The moisture retention curves for the horizons in the selected profiles were generated using three different methods:

2.2.2.1. Method A

Direct measurement of pressure head–moisture content relationships of the different horizons in the selected profile. Soil moisture characteristics were determined in the laboratory using standard desorption techniques (Hillel, 1980). Several 100 cm³ undisturbed core samples were taken from each soil horizon. Tension plate apparatus was used at suctions of 2, 10, 20, 50, 100, and 330 cm of water. For higher pressure values, i.e. 1, 3, and 15 bar, the pressure plate apparatus was used. For each soil sample the van Genuchten model (Eqs. 1, 2) was fitted to the observed $\theta-h$ data (Bork and Rohdenburg, 1986). As a result, a set of parameters (θ_s , θ_r , α , n) was available for every soil horizon of the different profiles. These parameter sets were subsequently used as input to the SWATRER model.

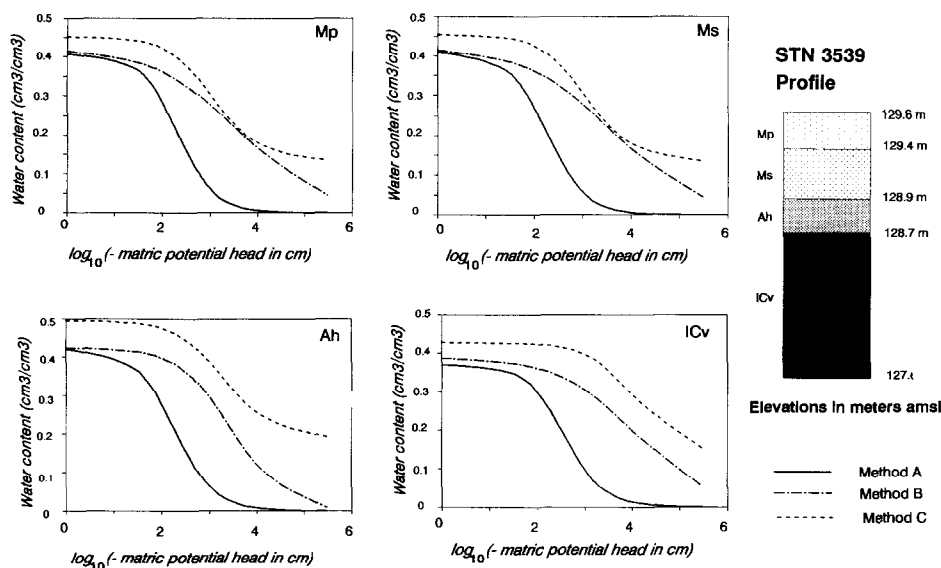


Fig. 1. Soil profile used in the comparison of estimation methods. Retention functions obtained with the three estimation methods are also shown for four soil horizons.

2.2.2.2. Method B

Measured hydraulic functions obtained after averaging over similar horizons. Replicate direct measurements obtained from all five profiles pertaining to a similar horizon description were combined into one group. The observed $\theta-h$ data was used to estimate field-scale mean parameters of the van Genuchten model using the non-linear optimization code RETC (van Genuchten et al., 1991). In this way we established field-scale mean values of the parameters of the moisture retention curve for all horizons.

2.2.2.3. Method C

Use of pedotransfer functions to predict the moisture retention characteristic. To estimate the retention curve $\theta(h)$ for a particular soil layer, the regression equations proposed by Vereecken et al. (1989) were used. Given the soil texture, carbon content and bulk density, the parameters of the van Genuchten model were estimated according to Eqs. 3(a)–3(d).

Due to the absence of direct measurements of saturated and unsaturated hydraulic conductivity the authors were forced to use pedotransfer functions (Eqs. 5(a)–(c)) for the estimation of the parameters of the Gardner model (Eq. 4) in all simulation runs. In other words, the three methods A, B, and C have identical $K(h)$ but different $\theta(h)$ functions.

2.2.3. Estimating other model inputs

In addition to soil hydraulic parameters, the accuracy of water balance simulation models is also dependent on the quality of other model input data. The prediction accuracy of water balance models relies on the successful characterization of the climatic, soil, and crop conditions prevailing throughout the simulation period. These values were fixed for all

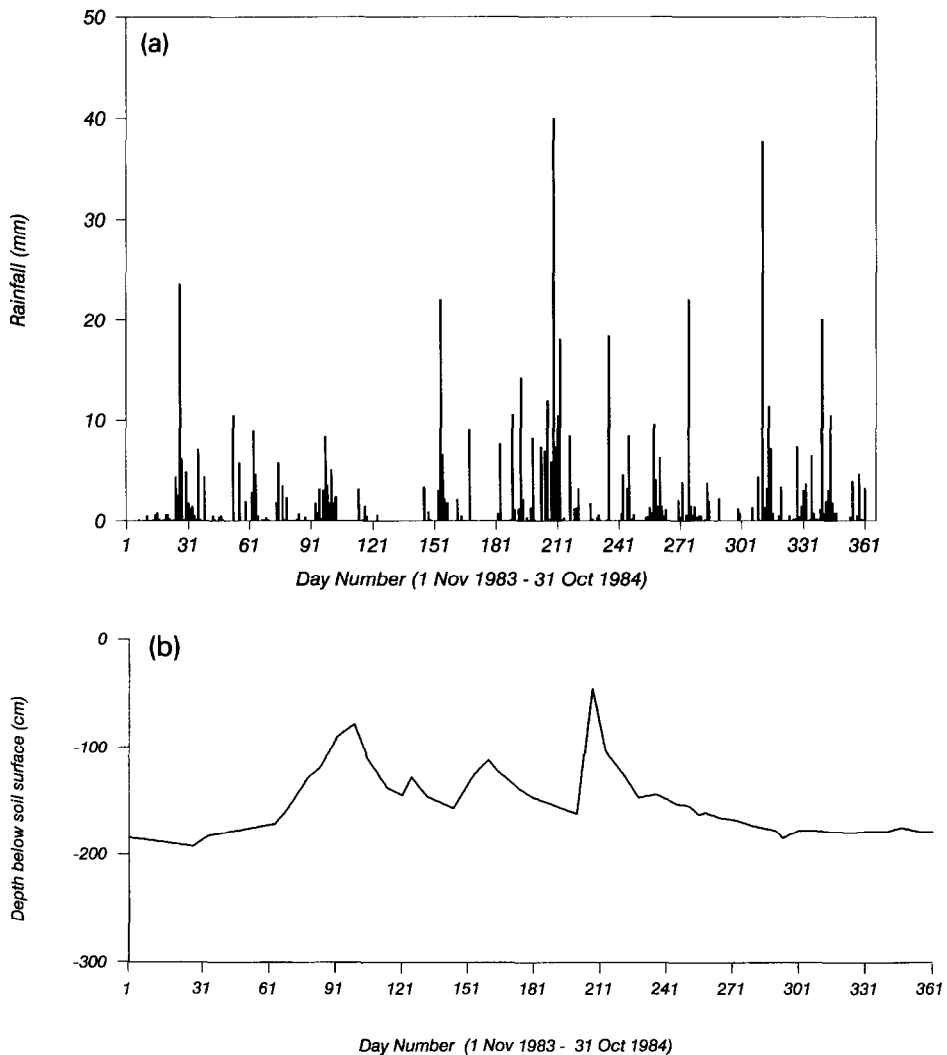


Fig. 2. Top and bottom boundary conditions used in the simulations. (a) Daily rainfall values, and (b) groundwater levels at soil profile STN 3539.

simulations. The simulation period started from November 1, 1983, until October 31, 1984.

Daily values of rainfall were derived from hourly measurements at the agrometeorological station located in the catchment area. The rainfall time series during the simulation period is presented in Fig. 2(a). Potential evapotranspiration data were taken from the BRG data set wherein the published values were derived from simulations using the AMBETI model (Braden, 1989). Groundwater levels were measured at odd intervals. Linear interpolation was performed to derive daily values of groundwater levels which were then input to the SWATRER model and used as a lower boundary condition (see Fig. 2(b)).

Table 2
Estimated moisture retention parameters for profile STN 3539

Layer	Horizon	Thickness (cm)	θ_r (cm ³ /cm ³)	θ_s (cm ³ /cm ³)	α (cm ⁻¹)	n
Method A						
1	Mp	20	0.00	0.4066	0.00474	1.2134
2	Ms	50	0.00	0.4090	0.00551	1.1559
3	Ah	20	0.00	0.4206	0.00532	1.1016
4	Icv	110	0.00	0.3704	0.00262	1.1501
Method B						
1	Mp	20	0.00	0.4210	0.00025	0.4920
2	Ms	50	0.00	0.4210	0.00025	0.4920
3	Ah	20	0.00	0.4250	0.00035	0.8350
4	Icv	110	0.00	0.3890	0.00009	0.5500
Method C						
1	Mp	20	0.1330	0.4520	0.00084	0.9530
2	Ms	50	0.1430	0.4550	0.00080	0.9470
3	Ah	20	0.1910	0.4960	0.00050	0.9360
4	Icv	110	0.1420	0.4280	0.00009	0.9360

Sugar beet was grown on the field during the simulation period. The different plant growth stages and leaf area index (LAI) were recorded by the BRG from sowing to harvest. Crop coefficients reducing potential evapotranspiration to actual evapotranspiration were taken from Doorenbos and Pruitt (1977). The rooting depth as a function of time was calculated according to the empirical description of Borg and Grimes (1986).

Simulations with SWATRER were performed with daily output of the state variables moisture content, pressure head, and drainage fluxes. Simulation results on moisture content were obtained at five soil depths (20, 40, 60, 80 and 100 cm below surface). Results for suction head were output at those soil depths where measured values were available, i.e. at 20, 45, 71, and 83 cm below surface. The monthly Darcian flux was calculated at 100 cm below surface. In the following sections we will focus our discussion on a subset of the model output, viz. observation depths 10–20, 50–60, and 90–100 cm for water content and 45 and 83 cm for pressure head.

3. Results and discussion

Simulation runs using SWATRER revealed that similar results were obtained for five profiles (Espino, 1992). Therefore, presentation of results will focus on one profile, i.e. profile STN 3539.

3.1. Comparison between moisture retention characteristics obtained through different methods

Parameter values for the moisture retention curves for each profile are presented in Table 2. The moisture retention curves obtained using Method A were similar for the three

Table

Table 3

Estimated parameters of the hydraulic conductivity relationship for profile STN 3539

Layer	Horizon	Thickness (cm)	K_s (cm day ⁻¹)	b (cm ⁻¹)	n'
1	Mp	20	303.9	1.318	1.543
2	Ms	50	274.6	1.480	1.513
3	Ah	20	480.5	1.867	1.464
4	Icv	110	169.9	1.867	1.464

upper layers. The bottom layer yielded quite different θ_s and α values. The residual moisture content, θ_r , for each of the layers estimated with Method A was zero, while the n parameter had a value greater than unity. Determination of the moisture retention curve using classical desorption techniques is a classical and widely accepted method (Hillel, 1980). Nevertheless, one should remember that the equilibration time to reach a zero water flux to the pressure plates increases as the soil cores become drier. Therefore, in practice, equilibration times are taken long enough to avoid this problem, although we realize that for the higher pressure ranges (starting from approximately pF 2.3), the observed moisture content may still be larger than the true one corresponding to a given pressure head. This may result in non-optimal measured retention curves. In other words, the transition from laboratory-scale measurements to field-scale applications may thus add another factor of uncertainty to our analysis. How large this uncertainty is will not be investigated here because it is beyond the scope of this study.

Unlike Method A, the n parameter using Method B yielded values less than one. The θ_s values calculated according to Method B are higher than Method A but lower compared to Method C.

Calculations of the θ_s and θ_r parameters using Method C resulted in higher values compared to the parameters obtained with Method A and B. The n parameter yielded values less than unity. For this particular profile, the estimated moisture retention curves of the different layers are clearly different from each other (see also Fig. 1). Table 3 shows the parameters of the Gardner equation estimated by Eqs. (5(a)–(c)).

3.2. Graphical comparison of simulated and measured moisture content and pressure head

For each profile, three simulation runs were made using sets of soil hydraulic parameters derived from the three methods discussed previously. Time series of measured moisture contents were compared to simulated values which were representative for the layers 10–20, 30–40, 50–60, 70–80 and 90–100 cm. Figs. 3–5 show results for three layers, i.e. 10–20, 50–60, and 90–100 cm. Similar results were obtained for the other two layers (not shown here). Near to the soil surface (see Fig. 3), the variations in the measured values are large with large differences between simulated and observed values. Deeper in the soil profile (Fig. 4 and Fig. 5), the variability in both observed and predicted moisture content decreases and there is a closer agreement between model calculations (using method A) and the observations. At the first observation depth (10–20 cm), the simulated values using

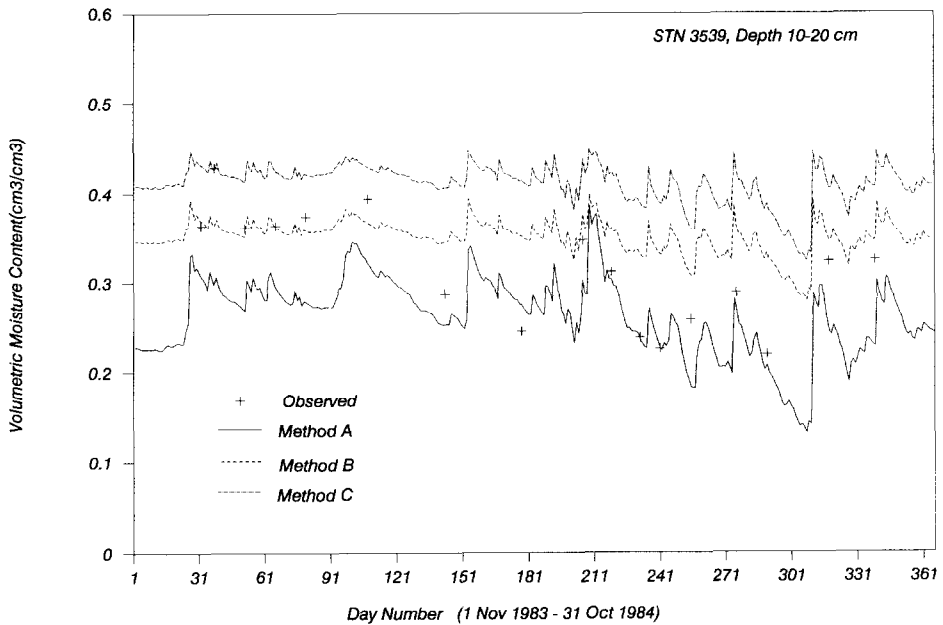


Fig. 3. Simulated and measured moisture content for profile STN 3539, 10–20 cm depth.

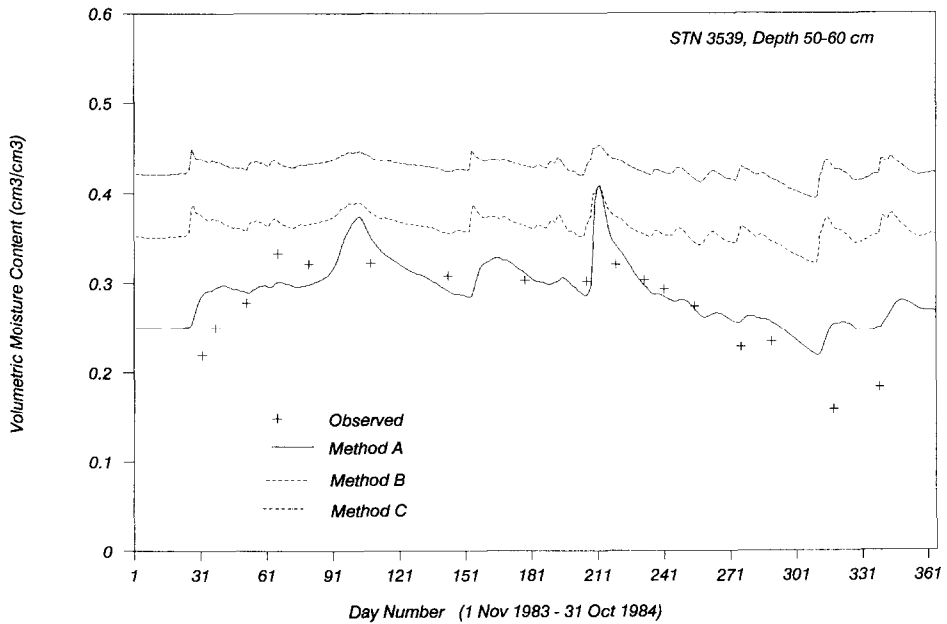


Fig. 4. Simulated and measured moisture content for profile STN 3539, 50–60 cm depth.

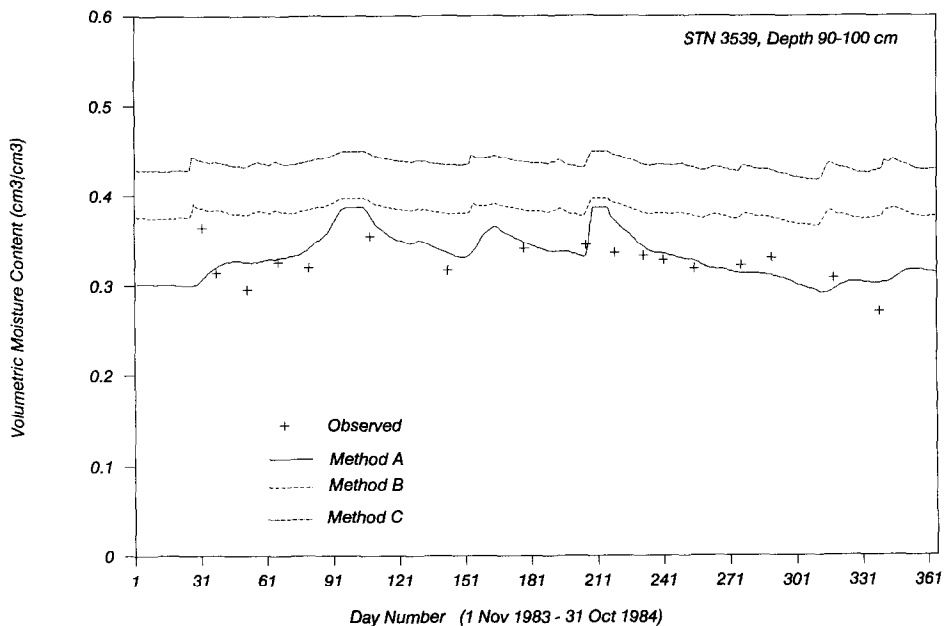


Fig. 5. Simulated and measured moisture content for profile STN 3539, 90–100 cm depth.

Method A are generally in closer agreement with the measured values, except between days 31 and 121 which correspond to the months December to February. Predictions at the time of the growing season (from day 151 to 331) are very close to the observed values. Simulations performed with average retention functions (Method B) showed the opposite behaviour: a close agreement when the soil was bare and an overprediction of the soil water content at the time of the growing season. Finally, predictions based on the pedotransfer functions (Method C) always overestimated the measured moisture contents. At deeper depths (Fig. 4 and Fig. 5), both Method B and C overpredict the measurements. Predictions based on Method A are now in agreement with the observations and this for the whole simulation period.

The simulations were done with a numerical model which has been tested on many sites for many different boundary conditions (e.g., Diels, 1994). However, the model is not perfect and as a result, discrepancies between observed and predicted values may be due to model deficiency in addition to improper hydraulic functions. Especially near the soil surface where effects of tillage on the soil water balance can be large, it is believed that the observed discrepancies before the start of the growing season are caused by an improper model representation of the hydraulic behaviour of an uncropped soil. More in particular, the difficulties related to the modelling of transpiration from a bare soil which has undergone several tillage practices make an interpretation of the simulation results rather difficult. Deeper in the profile these boundary effects are less pronounced and the comparison becomes more straightforward.

Another important variable when dealing with flow processes in soils is the soil water pressure head. Therefore, a comparison between simulated and measured values of pressure

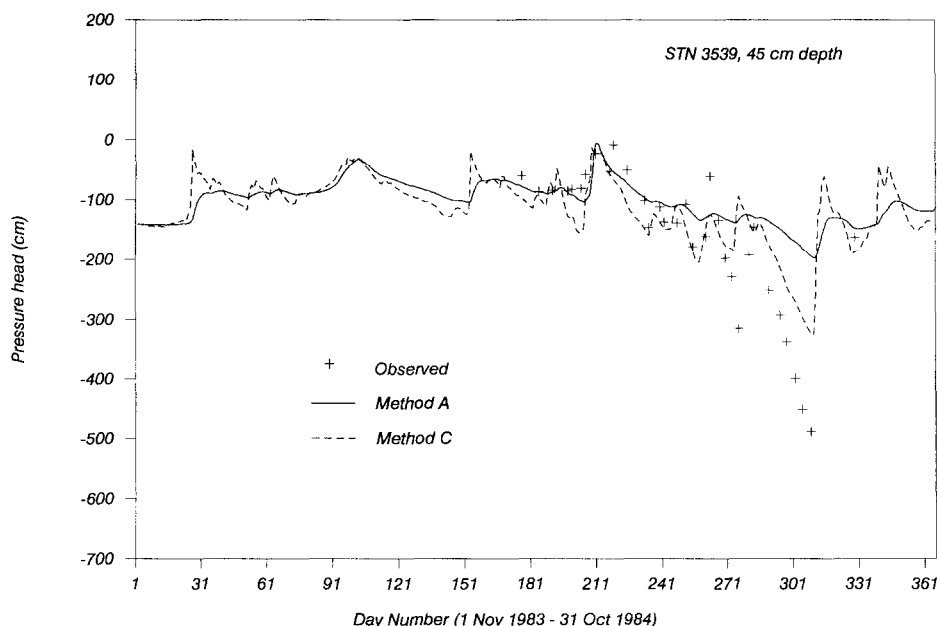


Fig. 6. Simulated and measured pressure head for profile STN 3539, 45 cm depth.

heads was conducted for Method A and C. Method B was omitted because it became already clear from the above discussion that it performed somewhere in between Method A and C. Within the simulation period, measurements of pressure heads were taken from the months of April to September resulting in 33 observations. Fig. 6 and Fig. 7 show the time series of predicted (Method A and C) and observed pressure head at a depth of 45 and 83 cm, respectively. Temporal fluctuations in simulated pressure heads using the MRC obtained from PTFs (Method C) are more pronounced compared to the predictions based on observed MRC (Method A). This is due to a flatter slope of the former and is also due to higher values of θ_s and θ_r compared to those pertaining to the latter. Hence, given a similar change in moisture content, the change in pressure head as seen from the MRC derived from method C will be definitely higher than the change shown in the MRC derived from Method A.

3.3. Statistical evaluation of simulated time series

Though it is evident that the use of Methods B and C resulted in a poor performance of the SWATRER model (Figs. 4–6), statistical analysis was conducted in one of the profiles (STN 3539) to have a quantitative comparison of the different estimation methods. Several statistical criteria were used to evaluate the performance of the three methods. These were: (1) maximum error (ME), (2) root mean square error (RMSE), (3) coefficient of determination (CD), (4) modelling efficiency (EF), and (5) coefficient of residual mass (CRM) (see also Loague and Green, 1991):

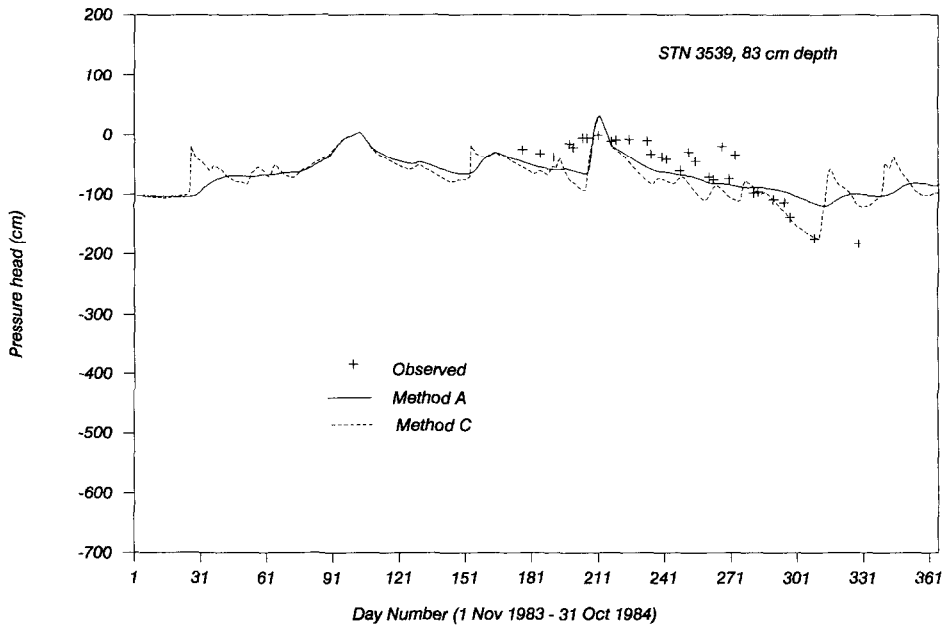


Fig. 7. Simulated and measured pressure head for profile STN 3539, 83 cm depth.

$$ME = \text{MAX}_{i=1}^n [ABS(P_i - O_i)] \quad (7)$$

$$RMSE = \left[\sum_{i=1}^n (P_i - O_i)^2 / n \right]^{1/2} \cdot (100 O_{\text{avg}}) \quad (8)$$

$$CD = \frac{\sum_{i=1}^n (O_i - O_{\text{avg}})^2}{\sum_{i=1}^n (P_i - O_{\text{avg}})^2} \quad (9)$$

$$EF = \frac{\sum_{i=1}^n (O_i - O_{\text{avg}})^2 - \sum_{i=1}^n (P_i - O_i)^2}{\sum_{i=1}^n (O_i - O_{\text{avg}})^2} \quad (10)$$

$$CRM = \frac{\sum_{i=1}^n O_i - \sum_{i=1}^n P_i}{\sum_{i=1}^n O_i} \quad (11)$$

where O_i is observed values, P_i is simulated values, O_{avg} is arithmetic average of observed values, and n is number of observations. The ME statistic represents the largest difference

Table 4

Statistical comparison between predicted and observed soil moisture content. RMSE = root mean square error, CD = coefficient of determination, EF = modelling efficiency, CRM = coefficient of residual mass, and ME = maximum error. Best performing statistic is indicated with asterisk

Depth (cm)	Method	Statistics RMSE	CD	EF	CRM	ME	Rank
	Range	$[0 \rightarrow \infty]$	$[0 \rightarrow \infty]$	$[-\infty \rightarrow 1]$	$[-\infty \rightarrow \infty]$	$[0 \rightarrow \infty]$	$1 \rightarrow 3$
	Best value:	0	1	1	0	0	1
10–20	A	20.91	0.94 *	–0.68 *	–0.16	0.13	1.6 *
	B	19.72 *	2.34	0.57	–0.11 *	0.11 *	1.6 *
	C	34.12	0.39	–1.55	–0.30	0.17	2.8
50–60	A	14.14 *	2.39	0.58 *	–0.06 *	0.09 *	1.4 *
	B	37.52	0.31 *	–2.22	–0.33	0.20	1.8
	C	60.24	0.10	–8.72	–0.57	0.27	2.8
90–100	A	6.91 *	0.91 *	–0.10 *	–0.17 *	0.06 *	1.0 *
	B	18.54	0.13	–6.40	–0.18	0.10	2.0
	C	34.55	0.04	–26.52	–0.34	0.16	3.0

between an observed and a predicted value and is used to indicate the worst case performance. The RMSE statistic expresses how much the observations are overpredicted or underpredicted by the model in terms of the mean value of the observations (percentage wise). The CD statistic compares the variability in the observations with that in the predictions. The EF statistic indicates the degree to which the predictions give a better estimate of observations compared to the mean of the observations. The degree of underprediction or overprediction is given by the CRM statistic.

For each method of generating input data these five criteria were computed. For each depth and each statistic the best performing method was indicated and all methods received a rank number depending on their performance. A mean rank based on the five statistics was then calculated for each depth and used to select the best performing method. The results of the statistical analysis are summarized in Table 4 and Table 5 for, respectively, water content and pressure head. From Table 4 it is evident that Method A and B give, on average, a similar performance with respect to the prediction of the moisture content in the upper 20 cm. In the deeper layers, however, Method A becomes superior to the two other methods. Method C always performed worst. The evaluation of Method A and C in terms

Table 5

Statistical comparison between predicted and observed soil moisture content. For an explanation of the statistical parameters, see Table 4

Depth (cm)	Method	Statistics RMSE	CD	EF	CRM	ME	Rank
	Range	$[0 \rightarrow \infty]$	$[0 \rightarrow \infty]$	$[-\infty \rightarrow 1]$	$[-\infty \rightarrow \infty]$	$[0 \rightarrow \infty]$	$1 \rightarrow 2$
	Best value:	0	1	1	0	0	1
40–50	A	62.5	3.63	0.27	1.69 *	295	1.8
	C	43.9 *	3.21 *	0.64 *	1.92	217 *	1.2 *
80–90	A	19.9 *	2.52	0.44 *	–0.23 *	81.1	1.4 *
	C	21.7	1.17 *	0.34	–0.48	60.8 *	1.6

of predicted pressure heads is given in Table 5. At 45 cm depth, Method C is clearly superior to method A. Deeper in the profile, the reverse is true, although the differences are less pronounced compared to the former depth. This is due to a smaller range of the observed pressure heads at a depth of 83 cm. For drier conditions closer to the soil surface, the indirectly estimated retention functions perform better. Conversely, for wetter conditions at greater depths, measured retention functions on average give better predictions.

3.4. Drainage flux

Downward fluxes of water are very important in studies on solute transport because the working equation in predicting solute movement is a function of the calculated water fluxes in the variably saturated soil–root zone. As a consequence, reliable predictions of drainage fluxes are of utmost importance if one is interested to also predict the fate of chemicals released from the soil surface. Therefore, we made a comparison between the simulated Darcian flux at a depth of 100 cm below ground surface using the three estimation methods. The comparison as it is presented here should not be considered as a true evaluation of the different methods in terms of predicted fluxes since no observed fluxes were available. In most cases information on field-scale drainage fluxes will not be available and comparisons will have to rely on certain assumptions. The assumption we made here is that we can consider the simulated drainage fluxes using Method A as a reference value, although we cannot test the quality of it. The uncertainty about the quality of the predicted fluxes is further increased as a result of the unknown (and hence estimated) hydraulic conductivity relationship. However, we should not exaggerate the negative effects on predicted water balance terms by using such indirectly estimated $K(h)$ curves. This has been demonstrated in our simulations for the case where direct measurements of the moisture retention curve were used in combination with estimated $K(h)$ curves. The predicted water contents and pressure heads were generally in good agreement with the observations. Furthermore, if one is interested in the drainage over a longer period, say one or several years, then the storage capacity of the soil is the most important factor, and not the hydraulic conductivity. Albeit, the comparison we made here remains a worthwhile exercise because it demonstrates the large effects of using an indirectly estimated water retention curve for the prediction of the drainage.

Fig. 8 shows monthly averaged Darcian fluxes for each estimation method over the entire simulation period. Positive fluxes indicate upward water movement as a result of capillary rise while negative fluxes reflect drainage. The direction of the calculated fluxes differed for the three methods during the months of March, May, June and October. It can also be seen in Fig. 8 that the difference between the simulated fluxes based on Method B and C is insignificant. The net annual cumulative flux below 100 cm depth was calculated for each of the three methods and amounted -32.3 mm for Method A, -119.3 mm for Method B, and -110.6 mm for Method C. Results indicate that the fluxes calculated using Method A gave the lowest drainage while fluxes obtained from the two other methods are almost four times as large as compared to Method A. This indicates that the calculated cumulative drainage based on PTFs would be generally higher than the simulation results using hydraulic parameters derived from direct laboratory measurements. One explanation for the observed differences could be that a smaller amount of water is required to saturate the soil

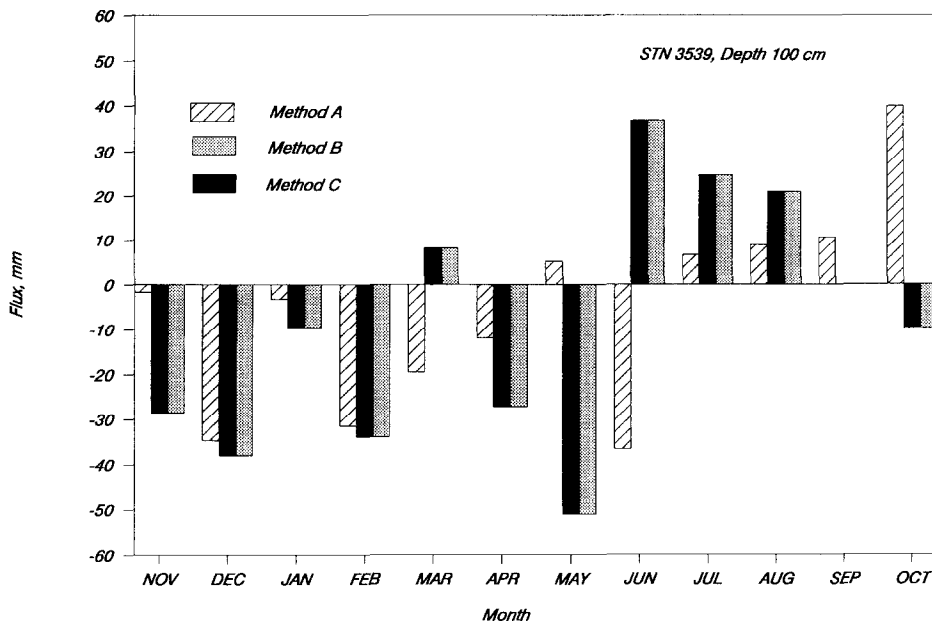


Fig. 8. Average monthly Darcian flux for profile STN 3539, 100 cm depth.

with MRCs that have a flatter slope such as the ones used here which were derived from pedotransfer functions. Also, smaller differences in water content between θ_s and θ_r (Method B and C compared to Method A) correspond to a smaller storage volume which leads to higher downward fluxes if Method B and C are used.

4. Conclusions and recommendations

Simulated water flow based on mechanistic models such as SWATRER are very sensitive to the parameters of the water retention characteristic and the hydraulic conductivity relationship. Soil hydraulic functions can be obtained either by direct measurement or estimated indirectly using PTFs. Methods of indirect estimation are rarely verified in terms of the hydraulic response of a cropped soil. In this study, we compared three methods for obtaining the soil water retention curve: (1) direct measurements with subsequent curve fitting using well established parametric models, (2) averaging of direct measurements over similar horizons resulting into mean retention functions, and (3) indirect estimation from readily available soil properties (percentage clay, silt, sand, etc.) using published pedotransfer functions (Vereecken et al., 1989; Vereecken et al., 1990). The comparison was based on the differences in the calculated moisture content, pressure head, and Darcian flux.

The shape of indirectly estimated retention curves differed significantly from measured retention curves, which was true for all horizons. Simulation results showed that the use of a soil moisture characteristic function derived from laboratory measurements (Method A)

gave better estimates of moisture content at deeper depths in the soil profile as compared to the two other methods. Closer to the soil surface, Method B was, on average, as good as Method A. Method C (PTFs) always gave the worst predictions. In other words, simulations using the pedotransfer functions in the form suggested by Vereecken et al. (1989); Vereecken et al. (1990) gave estimates of the moisture content that were 25 to 37% (on an annual base) in excess of the measurements. Predicted pressure heads at shallow depths (45 cm below ground surface) performed better when method C was used. Deeper in the soil profile, method A gave slightly better results compared to method C. This suggests that for drier conditions (higher pressure heads) method C is superior to method A. An additional reason for the relatively weak performance of the SWATRER model was the lack of measured data on unsaturated hydraulic conductivity.

In summary, the following cautionary notes on the use of PTFs should be made:

1. The use of PTFs is becoming increasingly popular but modellers should be careful about the use of these estimation methods knowing that they have been derived from limited information, and as a result, will never completely capture the hydraulic processes of interest. As suggested by Vereecken et al. (1989), improvements of the prediction equations can still be made by quantifying and incorporating the soil structure as a predictor variable. The difficulty here is how to quantify the soil structure in such a way that it yields useful predictor variables which can be combined with the other basic soil properties. One way to predict e.g. saturated hydraulic conductivity is by utilizing fractal principles (Rawls and Brakensiek, 1982). Another promising approach is the use of morphometric techniques to quantify, for instance, soil macroporosity. Booltink et al. (1993) and Hatano and Booltink (1992) used Methylene Blue staining patterns to quantify soil macroporosity which was then used as a parameter to predict the drainage from a soil column. More advanced techniques such as Computer Assisted Tomography (CAT) reveal the three-dimensional arrangement of structure elements and may result in a better prediction of the hydraulic conductivity near saturation (see for instance Phogat and Aylmore, 1989). The impact of the presence and absence of soil structure on the shape of the water retention curve was presented by Croney and Coleman (1954), Sharma and Uehara (1968), and Williams et al. (1990), among others.
2. Limitations of PTFs in general should be defined. They may be site specific or applicable only to a particular range of soil types from where the PTFs have been determined (one could give ranges of particle sizes for which they may be useful). Applications of pedotransfer functions to soils different from the ones used to derive PTFs, therefore, should be done with caution.
3. It is recommended that the performance of PTFs should first be evaluated on the soil series wherein these functions were derived in a similar manner as was presented here. In other words, testing the performance should be done in terms of specific applications, ranging from the prediction of state variables such as soil water content and pressure head to variables such as water deficit, water fluxes, trafficability.
4. Although we considered only one type of transfer function in our analysis (those of Vereecken et al., 1989; Vereecken et al., 1990), many alternative models exist to estimate parameters for hydraulic functions, and they should be included in a testing procedure prior to their application. The testing could be done based on a limited number of measured retention and conductivity curves collected from the site of interest. If detailed information is available on the hydraulic response of the soil, then the testing should be extended to this type of data.

5. An alternative to the type of transfer functions used here could be to perform measurements of $K(h)$ and $\theta(h)$ in well defined horizons of well defined soil series, which are more than only texture determined. The measured hydraulic functions can then be extrapolated to similar soil horizons, rather than estimating them from local soil properties. A good example is the so-called Staring Series (Wo 6\$ psten et al., 1986) which is based on measured $K(h)$ curves which are associated with soil horizon/texture classes as used in soil survey. Because the soil hydraulic functions are related to soil horizons and texture classes, they are called class-pedotransfer functions (Bouma, 1989). Satisfactory results obtained with class-pedotransfer functions were reported by Wo 6\$ psten et al. (1990)
6. With all the uncertainties involved, we first have to look for evidence that will give us confidence that using PTFs will not jeopardize our modelling results. Therefore, the effectiveness of pedotransfer functions to estimate the parameters of the hydraulic functions and their subsequent usefulness as input to water balance models should be further evaluated. In this evaluation one could also incorporate the prediction uncertainty by using Monte-Carlo analysis. In this way the observed variability in the input parameters is translated into uncertainties (confidence intervals) of the predicted values.

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